

Research of methods for modeling user assessments of 3D content for extended reality systems.

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Abstract—In this article, we will analyze methods for modeling user ratings of a video sequence and their potential application to extended reality systems. The main approaches and algorithms used to evaluate the quality of a video sequence are considered, their advantages and disadvantages for 2D and 3D content are revealed. The aim of the work is to develop our own method for evaluating the quality of 3D models to improve relevance and efficiency, taking into account the features of 3D content, since most of the metrics were created to meet the needs of two-dimensional systems. The research results can be used in the development and optimization of augmented reality systems, as well as in other areas where it is necessary to evaluate the quality of 3D models based on user preferences.

Keywords— *three-dimensional models; computer graphics; augmented reality; modeling user assessments; quality of 3D objects.*

I. INTRODUCTION

In today's world, extended reality (XR) technologies are becoming more popular and in demand. They are used in various fields such as education, medicine, manufacturing, entertainment, and others. One of the key tasks in the development of augmented reality systems is to create effective methods for user evaluation of video sequences. This is necessary to ensure high-quality and realistic representation of virtual objects on real images.

However, existing user assessment methods have a number of features that limit their use in augmented reality systems.

These include:

1. Evaluation bias;

Existing development methods were based on hypotheses about user behavior when watching a video sequence, rather than on virtual (3D) content. This can lead to biased conclusions that don't always match the real picture.

2. Low information content of metrics for 3D content.

Existing metrics apply mathematical analysis to a very limited extent, which is clearly insufficient for the features of 3D content and its perception by the human visual system.

User assessment methods to improve the quality and efficiency of augmented reality systems are an urgent task. The creation of such methods will improve users' perception

of virtual objects and increase loyalty to XR systems in general.

In the course of the research, we will examine the applicability of existing methods for modeling user ratings and try to create a more effective metric relevant to XR systems.

To achieve this goal, it is necessary to solve the following tasks:

1. Analyze existing user assessment methods;
2. To develop a new method that takes into account the features of the RNG and increases the accuracy and objectivity of estimates.

II. ANALYSIS OF EXISTING METHODS FOR MODELING USER RATINGS

It An overview of methods for evaluating two-dimensional images and video sequences

Video quality is a characteristic of the processed (editing, bitrate modification, encoding, decoding, compression) video, usually compared with the original [2].

There are two methods for evaluating (determining) video quality: subjective and objective.

The subjective assessment of quality is based on the opinion of viewers, while the objective assessment is based on algorithmic models that mimic subjective methods.

Two approaches are used to develop objective methods. The first one simulates the results of subjective user ratings (simulating the graphical display of the results). The second is the development of a predictive algorithm for user ratings based on the RNG and the features of the content itself.

Objective models can be classified depending on the amount of information available about the source signal: full-reference (FR), short-reference (RR), non-reference (NR), pixel (NR-P), parametric/bit-streaming (NR-B) and hybrid (Hybrid NR-P-B).

We will look at them in detail later.

Subjective assessment of quality

The subjective method of evaluating the quality is to receive feedback from viewers evaluating the video.

Initially, this method was used to evaluate the sound quality.

To conduct a subjective study, the following conditions must be met:

1. Select a video sequence for testing. As a rule, video fragments lasting about 5-10 seconds are used in order not to divert the attention of reviewers and reduce the total time spent on conducting experiments.;
2. Determine the research objectives and variable parameters (bitrate, speed, color range) of the video for comparison;
3. Select the test method;
4. Attract the necessary number of experts;
5. Analyze the results obtained based on their evaluation.

Subjective tests based on various recommendations (ITU-R BT.500 and others) are currently being conducted, for example, a subjective assessment of the quality of the new AV1 or H.266 encoding standard, and the subjective quality of Internet video codecs. Extensive subjective testing is conducted by VQEG (a group of Video Quality Experts).

In 1974, the first version of the recommendations was published, ITU-R BT.500 "Methodology for the subjective assessment of the quality of television images". These recommendations provide a comprehensive description of how to solve all of the above tasks.

Perception factors:

- 1) spatial accuracy (how clearly different parts of the scene are visible, are there any obvious distortions);
- 2) Temporal accuracy (how natural does the movement of the scene look, is it "smooth" and "smoothed out");
- 3) The person's environment, his inner state, the degree of involvement in interacting with the visual scene.

The most commonly used procedure from this standard is the double continuous quality scale method, in which the reviewer is shown pairs of short video clips A and B, one after the other, and asked to give fragments A and B a "quality rating" on a five-point scale from "Excellent" to "Poor". In each pair of sequences, one is an intact "reference" video sequence, and the other is the same sequence, but transformed by the video system under test.

Such a quality assessment requires repeated repetition with the participation of a large number of inexperienced reviewers, which makes it quite lengthy and expensive [2].

Absolute Category Rating (ACR)

Absolute category rating method (single source method) – According to this method, after each separately demonstrated test video sequence, viewers * are asked to rate the quality. If several viewers are simultaneously working with a video recording, the fragment should not exceed 10 seconds.

The following five-point scale should be used to evaluate the overall quality:

- 5 excellent;
- 4 good;
- 3 satisfactory;
- 2 unsatisfactory;
- 1 is bad.

Objective methods

Objective video quality assessment methods are algorithmic models that accurately mimic the work of subjective video quality assessment methods.

Objective models can be classified depending on the amount of information available about the original signal, the received signal, or whether the signal is present at all:

1. Full Reference Methods: FR models determine the quality difference by comparing the original video sequence with the received (processed) one.

As a rule, each pixel of the source is compared with the corresponding pixel in the processed video in terms of coordinates (relative to the length and width of the frame), without taking into account the intermediate state, the method of encoding or decoding, transmission or reception. FR models are usually the most accurate because they use the source video. Applicable only if the source video is available. The method is also characterized by high computational complexity, which increases linearly with increasing resolution and number of frames.

2. Abbreviated reference methods (RR):

The RR models extract some characteristics of both videos and compare them to get a quality score. They are used when the source video is unavailable or practically unrealizable, for example, when transmitting with limited bandwidth. RR is slightly inferior to FR with fewer inputs.

3. Non-reference methods NR (No-Reference Methods):

The NR models evaluate the quality of the processed video without using the original video sequence. Because of this, NR models may be less accurate than FR or RR models, but simpler to calculate.

- 3.1. Pixel methods NR-P (pixel-based methods):

Pixel-based models evaluate the decoded video sequence and analyze the quality using pixel information.

Some of these methods focus only on certain types of distortion. For example, on blurring or encoding artifacts.

- 3.2. Parametric/Bitstream Methods (NR-B):

NR-B analyzes the parameters extracted from the transmission container or the bit stream of the video sequence. For example, MPEG-TS packet headers, motion vectors, and quantization parameters. They do not require access to the original signal and do not need to be decoded, which greatly simplifies working with these methods and speeds up their work compared to those described above.

Since these methods do not have access to the decoded sequence, the accuracy of the quality assessment may decrease [5].

3.3. Hybrid methods Hybrid NR-P-B:

Hybrid models use parameters from a stream of bits with a decoded signal when estimating [5].

They combine the NR-P and NR-B methods.

MSE

The standard deviation is an indicator of the spread of values of a random variable [3].

This indicator is calculated using the formula (1) [3] for two monochrome images I and K with a size of $m \times n$, one of which is considered a noisy approximation of the other.:

$$MSE = 1/mn \sum(m-1) \sum(n-1) |I(i, j) - K(i, j)|^2 \quad (1)$$

A large MSE value has a large spread of values in the presented set with the average value of the set; a small value, respectively, indicates that the values in the set are grouped around the average value [3].

PSNR

The most popular indicator is the peak signal-to-noise ratio, but the limitations of this method have led to numerous attempts to develop more complex indicators that would better match the judgments of viewers of the "live broadcast" (SSIM, VMAF, VIF, DLM) [2].

PSNR (2) is determined on a logarithmic scale and calculated based on the standard deviation MSE of the source images from the transformed video frame relative to the number $(2n-1)^2$ (the square of the maximum possible pixel value) [2]:

$$PSNR(Db) = 10 \lg(2^n - 1)^2 / MSE \quad (2)$$

Where n is the pixel bit depth or the number of bits per sample (usually 8 bits)

This value is constantly used when comparing compressed and uncompressed video sequences [2]. Values above 40 dB indicate high image quality, while anything below 30-40 dB indicates low.

The obvious disadvantage of PSNR is that for its calculation it is necessary to have both the original and the converted video sequence, but the source is not always available, and even if it is, it does not always have absolute quality. PSNR also weakly correlates with subjective quality indicators [1].

SSIM

The structural similarity index is one of the methods used to assess the similarity between two images. SSIM is a full matching method that evaluates quality based on the original image (uncompressed or without distortion). SSIM is the development of traditional methods such as PSNR and MSE.

A distinctive feature of the method from those mentioned earlier (MSE and PSNR) is the consideration of "error perception" by processing structural changes in information.

The idea is that pixels have a close relationship, especially when they are located close to each other in space [4].

The SSIM metric is designed for different window sizes (estimated area per image) (3). The difference between two x and y windows of the same size is $N \times N$ [4]:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (3)$$

Where $\mu_x\mu_y$ is the average of x, y

$\sigma_x^2\sigma_y^2$ is the variance of x, y

σ_{xy} is the covariance of x and y ,

L is the dynamic range of pixels ($2n - 1$, usually 255).

The above formula is used only to determine the brightness of the image, which is used to evaluate the quality. The resulting SSIM ranges from -1 to +1. The value of +1 is achieved only if the samples are completely authentic. As a rule, the metric is designed for an 8×8 pixel window [4].

Overview of methods for evaluating three-dimensional models

Currently, there are no full-fledged methods for assessing the quality of three-dimensional models. Existing models compare two or more models with each other only based on the absolute values of one parameter, for example, spatial coordinates.

Let's consider methods for comparing three-dimensional models.

Like the methods of evaluating the quality of video sequences, they can be subjective and objective.

The objective ones usually compare the grids (meshes) of the reference model with the one being compared, while the subjective ones turn to the professional expertise of the respondents.

Grid comparison method

Evaluates the deviation of the spatial values of one model from others, focusing on the coordinates of each specific point of the 3D model grid.

The indicator works as follows: a distance limit is set. The grid from top to bottom has a gradient from red to blue. Red is the maximum positive deviation. Green indicates a deviation equal to or close to zero. Blue is the maximum negative deviation.

The method is actively used for archaeological and paleontological purposes to create optimized and at the same time high-quality 3D models of exhibits. However, this method has a number of features. First, it considers only vertices and does not compare either faces or planes. Therefore, the grid comparison method will not reveal the artifacts that occur in these structures. Secondly, it won't work without a much higher quality model.

RCVS method

There is also the Ray Casting Visual Search (RCVS) method, which is almost similar to the grid comparison

method, but the comparison is not carried out in a straight line, but in a sphere.

Instead of voxelization and rendering, this method uses ray analysis. The model is placed inside a polyhedral icosahedron, from the corners of which rays are cast and their length (path) to the surface of the model is recorded. The lists of these paths are sorted and compared with each other. Sorting eliminates rotations, since for models of the same or similar geometry, the ray paths will match within the margin of error, but differ in order.

III. DEVELOPMENT OF A NEW PLER METHOD

For an adequate assessment of the quality of 3D models, a technique is needed that takes into account the shape, size, texture and integrity of the model, as well as the features of the RF.

During the development of this technique, a three-dimensional model was placed in a virtual sphere from the inside of which, at equal angular intervals, segments (samples) are projected in the direction of its geometric center (the number of samples depends on the density of the 3D model grid) forming a segment AO equal to the radius of the circle R (Figure 1). B is the point of the first intersection of the segment AO (hereinafter referred to as R) surfaces (faces) of the model. By comparing the difference between the set of segments AO (hereinafter L) of the reference model and the studied one, the quality of the three-dimensional model is evaluated. This metric will be referred to as PLER (the ratio of peak length to error).

Since this indicator is based on existing PSNRs and RCVS, it will be easily integrated into other assessment systems and will work successfully with 3D models.

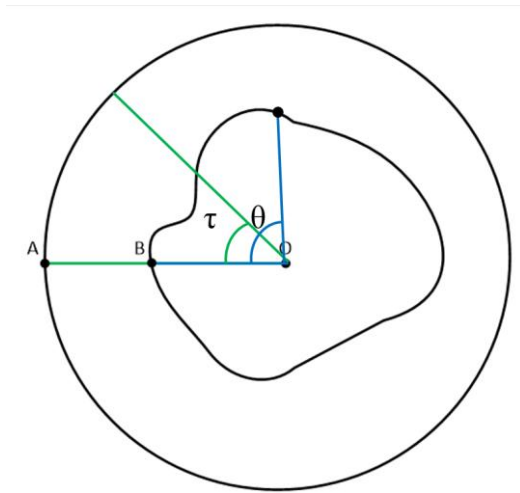


Fig. 1. the proposed PLER method

AO is a segment from the surface of the sphere to its center, equal to the radius of the sphere R.

AB is the segment plotted on the radius of the sphere from its surface to the surface of the 3D model, denoted by the letter L.

BO is a segment located on the radius of the sphere from the surface of the 3D model to the center of the sphere, denoted as S

$$AO = AB + BO \Rightarrow, \text{ hence } R = L + S$$

When evaluating the quality of models, the angular distance between the samples should be the same for both the reference and the studied model, and the orientation in space should match.

In addition, the two angular distances between the samples must be less than or equal to the angular distance between the two closest points in the reference model (4) (in accordance with Kotelnikov's theorem).

$$\theta \geq 2\tau \quad (4)$$

Where θ is the angle between the two closest points of the reference model,

τ - the angle between the samples

You can compare the objects being compared in space by searching for the maximum intersection of their volumes.

The numerical assessment of the quality of the three-dimensional model under study is determined on a logarithmic scale, since the viewer (and anyone in principle) perceives the 3D model using vision, and according to the Weber-Fechner law, which states that the intensity of perception of something, including images, linearly depends on the logarithm of the intensity of the stimulus. This metric is calculated based on the root-mean-square error MSE of the lengths L of the original 3D model from the L of the reference model:

$$PLER(dB) = 10 \lg \sum (R - L(\alpha, \beta, \gamma))^2 / MSE \quad (5)$$

$$MSE = \tau / (\alpha_{\max} \beta_{\max} \gamma_{\max}) \sum (\alpha_{\max}) \sum (\beta_{\max}) \sum (\gamma_{\max}) |I(\alpha, \beta, \gamma) - K(\alpha, \beta, \gamma)|^2 \quad (6)$$

Reducing the angular distance increases the accuracy of the indicator. The final value of the indicator can be considered correct if the decrease does not lead to a significant change in the value. A change in the value by more than 5% from the previous measurement is considered significant.

The final numerical value of the 3D model quality score is measured in dB.

It is important that both the reference and the three-dimensional models under study are in identical positions inside the sphere relative to each other.

IV. CONCLUSION

In the course of this work, methods for modeling user estimates of 2D images and video sequences, as well as 3D models, were investigated. The main approaches and algorithms used to evaluate the quality of 3D content in such systems were considered, and a proprietary metric for evaluating the quality of 3D models, PLER, was developed.

The developed method makes it possible to predict user preferences based on various characteristics of 3D models. PLEER takes into account parameters such as shape and size, and ensures high accuracy of forecasting with a high degree of probability.

Further research in this area may be aimed at creating more accurate and effective methods for modeling user ratings, as well as studying the influence of various parameters of 3D models on user preferences.

The PLEER method can be widely applied to create higher-quality and optimized 3D models in augmented reality systems.

REFERENCES

- [1] W. Stefan, "High-quality video", Wiley, March 2005, ISBN 0-470-02404-6.
- [2] V. Borovikov, "STATISTICS.", Description of the book on the website: For readers / V. Borisov., 688 p., St. Petersburg: Peter, 2003, ISBN 5-272-00078-1.
- [3] A. Loza and L. Mikhailovna, "Tracking objects based on structural similarity in video sequences", Proceedings of the 9th International Conference on Information Fusion, 2006.
- [4] M. Shahid, A. Rossholm, B. Levstrom and H.-J. Tsepertnik, "Image and video quality assessment without links: classification and review of recent approaches.", The Eurasian Journal of Image and Video Processing, 14 August 2014, doi: 10.1186/1687-5281-2014-40, ISSN 1687-5281.
- [5] N. Barman, Y. Reznik and M.G. Martini, "A subjective dataset for applications for streaming video from multiple screens.", arXiv: 2305.03138 [cs.MM], 2023.